Flight & Airline Database System (SQL + Python)

Module Title and Code: Introduction to Databases, UFCFTK-30-1

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TASK 2 (30%)

1. Consider the following **FLIGHT** table given below:

id	source	destination	aid	fare	discount	capacity
RA230	Kathmandu	Delhi	1	50000		200
AI215	Delhi	Kathmandu	2	60000	5000	300
3L046	Kathmandu	Abu Dhabi	3	70000		300
RA416	Kathmandu	Kuala Lumpur	1	55000	2000	180
H9891	Kuala Lumpur	Kathmandu	4	65000	3000	250

Similarly consider the following **AIRLINES** table given below:

id	name	country	founded_year	fleet_size
1	Air Nepal	Nepal	1990	20
2	Air India	India	1932	150
3	Etihad Airways	UAE	2003	120
4	Malaysia Airlines	Malaysia	1947	80

1.Write SQL syntax to create database with name **TRACKER** and the given tables: **FLIGHT** and **AIRLINES**. The **aid** column must refer to **id** of **AIRLINES** table.

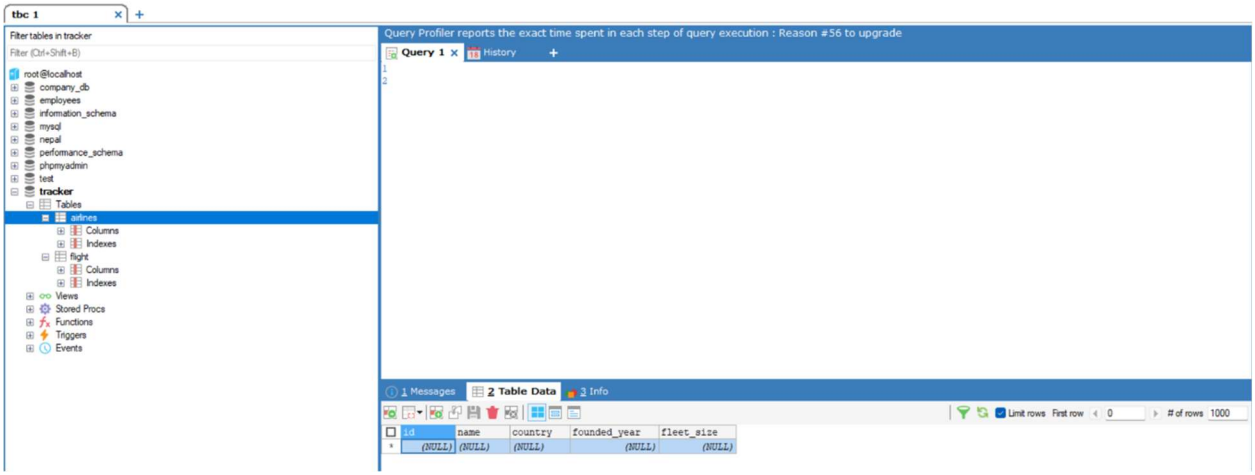
Answer:

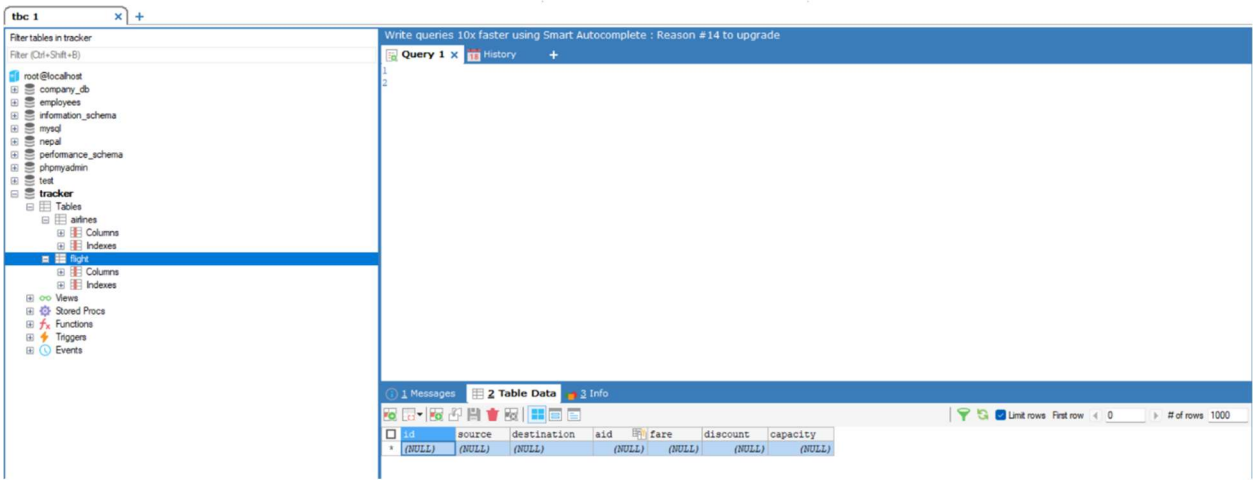
```
MariaDB [(none)]> CREATE DATABASE TRACKER;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> USE TRACKER;
Database changed
MariaDB [TRACKER]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| company_db         |
| employees          |
| information_schema |
| mysql              |
| nepal              |
| performance_schema |
| phpmyadmin         |
| test               |
| tracker            |
+--------------------+
9 rows in set (0.001 sec)

MariaDB [TRACKER]>
```

Firstly, the database **TRACKER** was created.

```
MariaDB [TRACKER]> CREATE TABLE AIRLINES (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(50) NOT NULL,
    ->     country VARCHAR(50) NOT NULL,
    ->     founded_year INT NOT NULL,
    ->     fleet_size INT NOT NULL
    -> );
Query OK, 0 rows affected (0.020 sec)

MariaDB [TRACKER]> CREATE TABLE FLIGHT (
    ->     id VARCHAR(10) PRIMARY KEY,
    ->     source VARCHAR(50) NOT NULL,
    ->     destination VARCHAR(50) NOT NULL,
    ->     aid INT NOT NULL,
    ->     fare DECIMAL(10, 2) NOT NULL,
    ->     discount DECIMAL(10, 2) DEFAULT NULL,
    ->     capacity INT NOT NULL,
    ->     FOREIGN KEY (aid) REFERENCES AIRLINES(id)
    -> );
Query OK, 0 rows affected (0.042 sec)
```

After that, the **AIRLINES** table was created first, followed by the **FLIGHT** table to ensure proper foreign key referencing (aid → AIRLINES.id).





Initially, both tables were empty.

2.Connect this database: **TRACKER** with Python and create a Python script that inserts at least 5 rows with random data into the FLIGHT and AIRLINES table.

Answer:

The TRACKER database was connected using Python in Jupyter Notebook (via VS Code). A Python script was executed to insert at least 5 rows into both the **FLIGHT** and **AIRLINES** tables.

```
import mysql.connector
print("Working")
```

[1] ✓ 0.1s

```
Working
```

```
def connect_database():
    try:
        conn = mysql.connector.connect(
            host="localhost",
            user="root",
            password="Raunak@9801",
            database="TRACKER"
        )
        print("Successfully connected to database!")
        return conn
    except mysql.connector.Error as err:
        print(f"Error: {err}")
        return None
```

[2] ✓ 0.0s

```python
def insert_sample_data(connection):
    try:
        cursor = connection.cursor()


        airlines_data = [
            (1, 'Air Nepal', 'Nepal', 1990, 20),
            (2, 'Air India', 'India', 1932, 150),
            (3, 'Etihad Airway', 'UAE', 2003, 120),
            (4, 'Malaysia Airlines', 'Malaysia', 1947, 80),
            (5, 'Qatar Airways', 'Qatar', 1993, 200)
        ]

        cursor.executemany(
            "INSERT INTO AIRLINES VALUES (%s, %s, %s, %s, %s)",
            airlines_data
        )


        flight_data = [
            ('RA230', 'Kathmandu', 'Delhi', 1, 50000, None, 200),
            ('AI215', 'Delhi', 'Kathmandu', 2, 60000, 5000, 300),
            ('3L046', 'Kathmandu', 'Abu Dhabi', 3, 70000, None, 300),
            ('RA416', 'Kathmandu', 'Kuala Lumpur', 1, 55000, 2000, 180),
            ('H9891', 'Kuala Lumpur', 'Kathmandu', 4, 65000, 3000, 250)
        ]

        cursor.executemany(
            """INSERT INTO FLIGHT
            (id, source, destination, aid, fare, discount, capacity)
            VALUES (%s, %s, %s, %s, %s, %s, %s)""",
            flight_data
        )
```

```python
        connection.commit()
        print("Successfully inserted 5 rows in each table!")

    except mysql.connector.Error as err:
        print(f"Error inserting data: {err}")
        connection.rollback()
    finally:
        if cursor:
            cursor.close()
```

✓ 0.0s

```python
if __name__ == "__main__":
    db_connection = connect_database()
    if db_connection:
        insert_sample_data(db_connection)
        db_connection.close()
        print("Database connection closed")
```

✓ 0.0s

```
Successfully connected to database!
Successfully inserted 5 rows in each table!
Database connection closed
```

After executing the script, the data was successfully inserted.

To verify the insertion, SQLyog was used to view the tables.

1 Messages | 2 Table Data | 3 Info

id	name	country	founded_year	fleet_size
1	Air Nepal	Nepal	1990	20
2	Air India	India	1932	150
3	Etihad Airway	UAE	2003	120
4	Malaysia Airlines	Malaysia	1947	80
5	Qatar Airways	Qatar	1993	200
(NULL)	(NULL)	(NULL)	(NULL)	(NULL)

1 Messages | 2 Table Data | 3 Info

id	source	destination	aid	fare	discount	capacity
3L046	Kathmandu	Abu Dhabi	3	70000.00	(NULL)	300
AI215	Delhi	Kathmandu	2	60000.00	5000.00	300
H9891	Kuala Lumpur	Kathmandu	4	65000.00	3000.00	250
RA230	Kathmandu	Delhi	1	50000.00	(NULL)	200
RA416	Kathmandu	Kuala Lumpur	1	55000.00	2000.00	180
(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)

3.Execute the following using SQLyog or any alternatives DBMS:

Answer:

a. SQL query to list all the flights of source to destination on which flight is capable to hold maximum passengers.

SELECT source, destination, MAX (capacity) AS maximum

FROM FLIGHT

GROUP BY source, destination

ORDER BY maximum DESC;

```
MariaDB [TRACKER]> SELECT source, destination, MAX(capacity) AS maximum
    -> FROM FLIGHT
    -> GROUP BY source, destination
    -> ORDER BY maximum DESC;
+--------------+--------------+---------+
| source       | destination  | maximum |
+--------------+--------------+---------+
| Delhi        | Kathmandu    |     300 |
| Kathmandu    | Abu Dhabi    |     300 |
| Kuala Lumpur | Kathmandu    |     250 |
| Kathmandu    | Delhi        |     200 |
| Kathmandu    | Kuala Lumpur |     180 |
+--------------+--------------+---------+
5 rows in set (0.001 sec)
```

This query retrieves flights between each source and destination with the maximum passenger capacity.

b. SQL query to retrieve the total flights that don’t take off from Kathmandu.

SELECT COUNT (*) AS noFlightKTM

FROM FLIGHT WHERE source! = 'Kathmandu';

```
MariaDB [TRACKER]> SELECT COUNT(*) AS noFlightKTM
    -> FROM FLIGHT
    -> WHERE source != 'Kathmandu';
+-------------+
| noFlightKTM |
+-------------+
|           2 |
+-------------+
1 row in set (0.001 sec)
```

This query counts the number of flights that do not depart from Kathmandu.

c. SQL syntax to update the fare of all flights by 2% to those flights either capacity is more than 250 or take off from Kathmandu.

Current Data:

id	source	destination	aid	fare	discount	capacity
3L046	Kathmandu	Abu Dhabi	3	70000.00	(NULL)	300
AI215	Delhi	Kathmandu	2	60000.00	5000.00	300
H9891	Kuala Lumpur	Kathmandu	4	65000.00	3000.00	250
RA230	Kathmandu	Delhi	1	50000.00	(NULL)	200
RA416	Kathmandu	Kuala Lumpur	1	55000.00	2000.00	180
* (NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)

UPDATE FLIGHT

SET fare = fare * 1.02

WHERE capacity > 250

OR source = 'Kathmandu';

```
MariaDB [TRACKER]> UPDATE FLIGHT
    -> SET fare = fare * 1.02
    -> WHERE capacity > 250 OR source = 'Kathmandu';
Query OK, 4 rows affected (0.005 sec)
Rows matched: 4  Changed: 4  Warnings: 0
```

The query increases the fare by 2% for selected flights.

After Modification:

id	source	destination	aid	fare	discount	capacity
3L046	Kathmandu	Abu Dhabi	3	71400.00	(NULL)	300
AI215	Delhi	Kathmandu	2	61200.00	5000.00	300
H9891	Kuala Lumpur	Kathmandu	4	65000.00	3000.00	250
RA230	Kathmandu	Delhi	1	51000.00	(NULL)	200
RA416	Kathmandu	Kuala Lumpur	1	56100.00	2000.00	180
* (NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)

Multiple Times:

id	source	destination	aid	fare	discount	capacity
3L046	Kathmandu	Abu Dhabi	3	74284.56	(NULL)	300
AI215	Delhi	Kathmandu	2	63672.48	5000.00	300
H9891	Kuala Lumpur	Kathmandu	4	65000.00	3000.00	250
RA230	Kathmandu	Delhi	1	53060.40	(NULL)	200
RA416	Kathmandu	Kuala Lumpur	1	58366.44	2000.00	180
* (NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)

When the query was executed multiple times, different outputs were observed because each execution increased the fare again by 2% on the already updated value.

Mathematically:

After 1st run → fare × 1.02

After 2nd run → fare × $(1.02)^2$

After 3rd run → fare × $(1.02)^3$

Therefore, the query should be executed only once to maintain correct results.

d. SQL query to find the fare and capacity of all flights whose discount is not available with destination having at least 5 characters.

SELECT fare, capacity

FROM FLIGHT

WHERE discount IS NULL AND CHAR_LENGTH (destination) >= 5;

```
MariaDB [TRACKER]> SELECT fare, capacity
    -> FROM FLIGHT
    -> WHERE discount IS NULL
    -> AND CHAR_LENGTH(destination) >= 5;
+----------+----------+
| fare     | capacity |
+----------+----------+
| 74284.56 |      300 |
| 53060.40 |      200 |
+----------+----------+
2 rows in set (0.001 sec)
```

This query retrieves the fare and capacity of flights where no discount is available and the destination name has at least 5 characters.